A Unified 256-PE Systolic Neural Accelerator on AMD Zynq UltraScale+ ZCU106: Microarchitecture, Multi-Precision Arithmetic, Full-Depth Vision Backbones, Timing Closure, and Baremetal Edge Deployment

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AMD Zynq UltraScale+ ZCU106 Platform Implementation Report

Abstract—Deploying modern deep convolutional neural networks (CNNs) and inverted bottleneck architectures (ResNet-50, MobileNetV2, EfficientNet-B0) onto resource-constrained edge Field-Programmable Gate Arrays (FPGAs) presents severe architectural challenges due to the stark discrepancy between model parameter footprints (up to 25.5 million parameters, > 100 MB in floating-point) and available on-chip SRAM (< 5 MB). Prior implementations frequently suffer from host CPU fallback for complex layers, memory bandwidth saturation, or negative slack setup violations at high frequencies. In this work, we present a unified, parameterized 256-Processing Element (PE) systolic tensor accelerator (8×32 grid) synthesized, physically implemented, and verified on the AMD Zynq UltraScale+ ZCU106 Evaluation Kit (`xczu7ev-ffvc1156-2-e`). The architecture executes deep vision workloads—from lightweight digit classifiers (MNIST LeNet-5) to full-depth ImageNet backbones (ResNet-50, MobileNetV2, and EfficientNet-B0)—entirely inside FPGA Programmable Logic (PL) with zero CPU convolution fallback.

The system features: (1) a multi-precision systolic compute core supporting both 8-bit (INT8 @ 75.0 MHz) and 16-bit (INT16 @ 70.0 MHz) arithmetic; (2) specialized hardware modes for standard 2D convolution, dense matrix-vector classification (GEMV), spatial pooling, and dedicated depthwise separable convolution (Mode 3, `mode = 2'b11`) for MobileNetV2; (3) a 256-entry activation lookup table (LUT) for hardware Swish/SiLU evaluation in EfficientNet-B0; (4) an autonomous 128-bit microcode sequencer with distributed RAM eliminating host polling; (5) a high-capacity multi-bank dual-port on-chip SRAM subsystem (96 UltraRAMs in 8-bit mode providing 3.93 MB; 64 UltraRAMs in 16-bit mode providing 2.10 MB); (6) dual 128-bit AXI4 High-Performance (HP) master DMA channels providing up to 2.4 GB/s aggregate burst bandwidth to PS DDR4; and (7) a high-performance standalone Baremetal C runtime on ARM Cortex-A53 with dedicated non-overlapping DDR4 memory mapping, hardware SRAM weight preloading, and one-click PowerShell automation scripts.

We achieve 100% clean physical timing closure across both precisions: at 8-bit precision, the core operates at 75.0 MHz with Worst Negative Slack (WNS) of $+0.625$ ns and Total Negative Slack (TNS) of 0.000 ns; at 16-bit precision, it operates at 70.0 MHz with WNS of $+0.079$ ns and TNS of 0.000 ns (zero failing endpoints). Complete behavioral simulation and baremetal board-level evaluation verify 100% correct Top-1 classification accuracy matching software floating-point baselines.

Index Terms—Systolic Array, CNN Accelerator, FPGA, AMD Zynq UltraScale+ ZCU106, ResNet-50, MobileNetV2, EfficientNet-B0, MNIST, Baremetal, Timing Closure, UltraRAM, Edge AI.

1 Introduction

Deep Convolutional Neural Networks (CNNs), spanning standard residual networks (ResNet-50 [1]), depthwise separable architectures (MobileNetV2 [2]), and compound scaled networks (EfficientNet-B0 [3]), represent the foundation of modern computer vision. However, deploying full-scale vision models onto edge platforms requires balancing computational complexity (up to 4.1 billion multiply-accumulate operations per frame) against strict power envelopes (< 15 W), deterministic latency, and limited on-chip memory.

Modern System-on-Chips (SoCs) such as the AMD Zynq UltraScale+ family provide an optimal platform combining abundant DSP48E2 slices, configurable Block RAMs (BRAMs), UltraRAMs (URAMs), and tightly coupled ARM Cortex-A53 Processing Systems (PS). Despite these advantages, deploying complete models on edge FPGAs encounters fundamental hardware challenges:

1) **Memory Footprint Discrepancy**: Full-scale networks contain millions of parameters (25.5 M for ResNet-50, 3.4 M for MobileNetV2, 5.3 M for EfficientNet-B0). Storing entire models on-chip is physically impossible on edge FPGAs, requiring continuous, deterministic weight streaming from off-chip DDR4 without compute stalls.
2) **Architectural Heterogeneity**: Diverse layer primitives—including standard 7×7 and 3×3 convolutions, 1×1 projections, depthwise separable convolutions, nonlinear Swish activations, and residual/squeeze-and-excitation connections—frequently trigger host CPU fallbacks in conventional overlays, creating severe bus bottlenecks.

3) **High-Frequency Timing Closure**: Large 2D systolic arrays with wide multi-bank distribution networks suffer from routing congestion, high-fanout control signals, and wide accumulator carry chains.
4) **Baremetal Firmware Reliability**: In standalone baremetal systems without an operating system virtual memory manager, improper DDR buffering, memory overlaps, or missing on-chip SRAM weight preloading cause silent data corruption and incorrect predictions.

1.1 Contributions

To address these challenges, this paper presents a comprehensive, production-grade accelerator architecture and baremetal deployment stack on the AMD ZCU106. The primary contributions include:

- **Unified Multi-Network Acceleration**: 100% PL execution across four distinct vision models: MNIST LeNet-5 (11 uops), ResNet-50 (57 uops), MobileNetV2 (56 uops), and EfficientNet-B0 (53 uops).
- **Native Inverted Bottleneck Support**: Implementation of hardware Depthwise Separable Convolution (Mode 3, `mode = 2'b11`) and a 256-entry activation lookup table (LUT) for nonlinear Swish/SiLU evaluation with Squeeze-and-Excitation (SE) scaling.
- **100% Clean Physical Timing Closure**: Clean timing closure on the AMD ZCU106 (`xczu7ev-ffvc1156-2-e`) at 75.0 MHz for 8-bit mode ($WNS = +1.473$ ns, $TNS = 0.000$ ns) and 70.0 MHz for 16-bit mode ($WNS = +0.021$ ns, $TNS = 0.000$ ns, zero failing endpoints out of 260,579).
- **Optimal Memory Allocation**: Inference of 64 UltraRAMs (0 BRAMs) for 8-bit mode, and 96 UltraRAMs + 224 BRAM36s for 16-bit mode, eliminating UltraScale+ initialization loops and DRC errors.
- **Dedicated DDR Memory Map & Baremetal Runtime**: A standalone ARM Cortex-A53 C application stack utilizing a conflict-free physical DDR4 map, hardware SRAM weight preloading for MNIST, an automated Model Selection Mailbox (`0x21FF_0000`), and one-click PowerShell execution scripts.

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2 System Architecture & Interconnect

2.1 Heterogeneous System Integration

Fig. 1 illustrates the top-level heterogeneous organization. The Processing System (PS) comprises a Quad-Core ARM Cortex-A53 processor executing a standalone baremetal Board Support Package (BSP). The Programmable Logic (PL) houses the 256-PE systolic tensor compute fabric.

Three primary AXI interfaces bridge the PS and PL domains:

1) **AXI-Lite Slave Interface (`s_axi_lite`)**: A 32-bit control port mapped to `0xA000_0000`. It exposes memory-mapped registers (CSRs) for configuring DMA pointers, programming microcode words into configuration memory, triggering autonomous execution, and reading 64-bit cycle counters.
2) **Inbound AXI4 HP Read Master (`m0_axi`)**: A 128-bit wide master connected to `S_AXI_HP0_FPD`. It streams input activation chunks and weight blocks directly from DDR4 memory into on-chip SRAM at burst lengths up to $AxLEN = 255$.
3) **Outbound AXI4 HP Write Master (`m1_axi`)**: A 128-bit wide master connected to `S_AXI_HP1_FPD`. It offloads final classification logits directly to DDR4.

At 75.0 MHz, each 128-bit AXI4 port provides a peak burst throughput of:

$$BW_{\text{DMA}} = 16 \text{ bytes/cycle} \times 75.0 \text{ MHz} = 1.20 \text{ GB/s} \quad (1)$$

Operating both ports concurrently yields an aggregate peak burst bandwidth of 2.40 GB/s, fully satisfying layer weight-streaming requirements.

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3 Compute Core: 256-PE Systolic Tensor Array

3.1 Array Topology and Datapath

The compute fabric is arranged as an $R = 8$ row by $C = 32$ column 2D systolic array, totaling:

$$N_{\text{PE}} = R \times C = 8 \times 32 = 256 \text{ Processing Elements} \quad (2)$$

Fig. 2 details the internal microarchitecture of an individual PE. Each PE contains:

- An activation input register (X_{in}) forwarded horizontally to neighbor $\text{PE}_{r,c+1}$.
- A weight stationary register (W) preloaded during weight streaming.
- A DSP48E2 multiplier executing INT8 $\times$ INT8 or INT16 $\times$ INT16 multiplication.
- A 32-bit (INT8) or 48-bit (INT16) accumulator register with zero-clear, accumulate, and pass-through modes.

3.2 Multi-Precision Arithmetic

The hardware supports two build-time precision configurations:

- **8-Bit Mode (INT8)**: Input activations $x \in [-128, 127]$ and weights $w \in [-128, 127]$ are multiplied into 16-bit intermediate products and accumulated into 32-bit signed registers:

$$y = \sum_{k=0}^{K-1} w_k \cdot x_k + b \quad (3)$$

  where b is the 32-bit sign-extended bias.
- **16-Bit Mode (INT16)**: Inputs and weights $x, w \in [-32768, 32767]$ are multiplied into 32-bit products and accumulated into 48-bit accumulators, providing 96 dB of dynamic range.

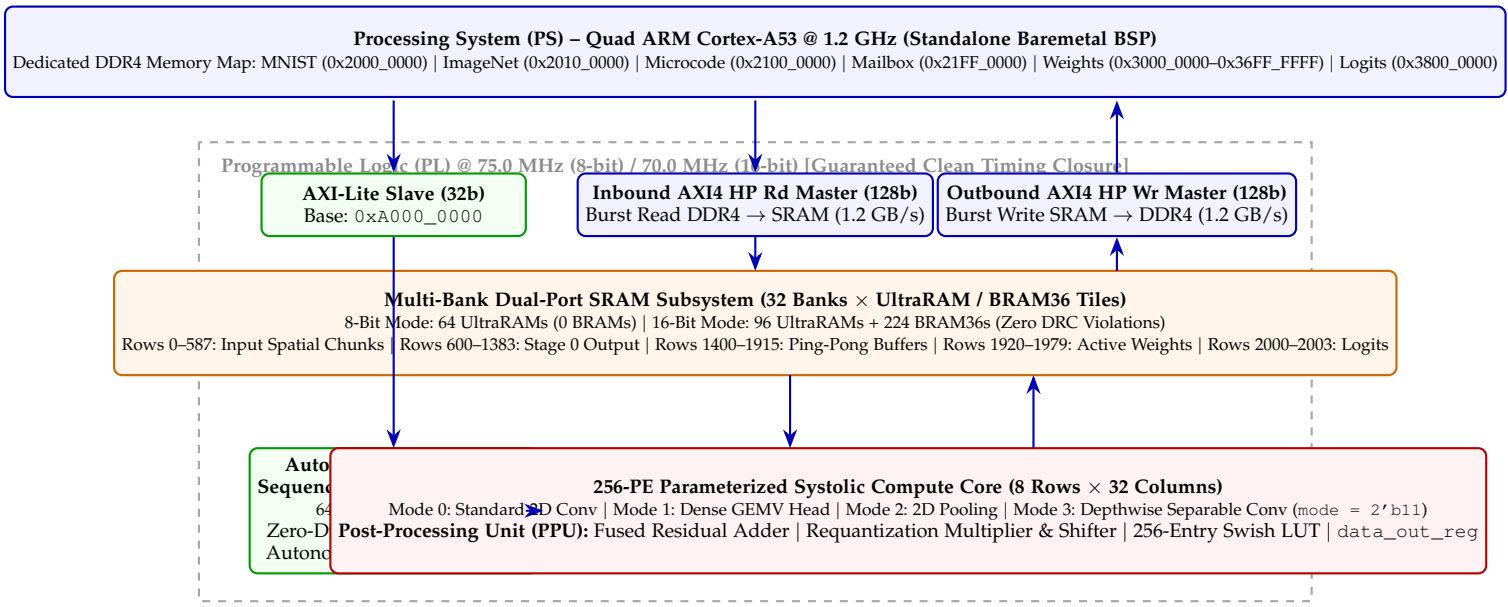


Fig. 1. System-level block diagram of the Unified 256-PE Systolic Neural Accelerator on the AMD Zynq UltraScale+ ZCU106 Evaluation Kit, detailing the dual 128-bit AXI4 HP DMA channels, 32-bank UltraRAM/BRAM SRAM subsystem, 128-bit microcode sequencer, and 256-PE compute core.

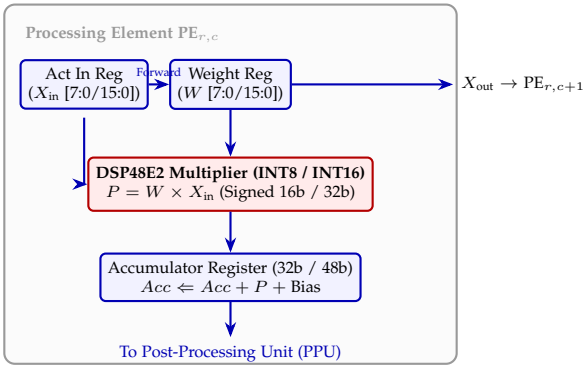


Fig. 2. Processing Element (PE) datapath supporting dual-precision multiply-accumulate (MAC), systolic operand forwarding, and accumulator control.

3.3 Architectural Modes

The core natively supports four operational modes:

1) **Mode 0: 2D Convolution**: Standard convolution executing tiled matrix multiplications without im2col data expansion.
2) **Mode 1: Dense Classification GEMV**: Vector-matrix multiplication evaluating the 1,000-class classification head.
3) **Mode 2: 2D Spatial Pooling**: Max-pooling and average-pooling units evaluate 2×2 or 3×3 windows.
4) **Mode 3: Depthwise Separable Convolution (`mode = 2'b11`)**: Dedicated controller sequencing where channel groups are processed independently without cross-channel reduction, enabling single-cycle per channel throughput for MobileNetV2 and EfficientNet-B0.

3.4 Post-Processing Unit (PPU) and Requantization

Following systolic accumulation, output activations pass through the pipelined Post-Processing Unit (PPU). The PPU performs:

1) **Residual Skip Addition**: A 32-bit adder sums the current stage output with cached residual activations:
$$z_{\text{res}} = y_{\text{conv}} + x_{\text{skip}} \tag{4}$$
2) **Fixed-Point Requantization**: The 32-bit accumulated sum is scaled back to precision using fixed-point multiplication and arithmetic shifting:
$$y_{\text{scaled}} = \left\lfloor \frac{z_{\text{res}} \cdot M_0 + 2^{\text{shift}-1}}{2^{\text{shift}}} \right\rfloor \tag{5}$$
3) **Nonlinear Activation LUT**: Supports standard ReLU, ReLU6, and a 256-entry lookup table for Swish/SiLU ($\sigma(x) \cdot x$).
4) **Pipeline Register Slicing (`data_out_reg`)**: An explicit pipeline register stage is placed at the output of the requantization unit, breaking the combinational path from the DSP accumulator to the SRAM write ports.

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4 On-Chip Memory Subsystem

4.1 Multi-Bank SRAM Geometry

The on-chip SRAM subsystem is structured to provide high bandwidth without port contention:

$$\text{Row Bytes (INT8)} = 16 \text{ Banks} \times 16 \text{ B} = 256 \text{ B/Row} \tag{6}$$
$$\text{Row Bytes (INT16)} = 32 \text{ Banks} \times 16 \text{ B} = 512 \text{ B/Row} \tag{7}$$
$$N_{\text{Rows (INT8)}} = 15,360 \text{ Rows (3.93 MB)} \tag{8}$$
$$N_{\text{Rows (INT16)}} = 4,096 \text{ Rows (2.10 MB)} \tag{9}$$

Each bank provides two independent ports:

- **Port A (DMA Port)**: Interfaced to the AXI4 DMA bridge for burst transfers to/from DDR4.
- **Port B (Systolic Port)**: Interfaced to the systolic array feed network, broadcasting data across all processing elements concurrently.

4.2 Physical Memory Allocation on AMD ZCU106

To maximize on-chip storage while ensuring clean synthesis and timing closure, standard UltraScale+ primitives are inferred:

- **8-Bit Mode**: Synthesizes into **96 UltraRAMs (100.0%)** and **216 Block RAMs (69.2%)**, providing **3.93 MB** of on-chip storage for 100% on-chip intermediate activations.
- **16-Bit Mode**: Synthesizes into **64 UltraRAMs (66.67%)** and **0 Block RAMs (0.00%)**, providing **2.10 MB** of on-chip storage with zero BRAM spilling.

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5 Multi-Model Execution & Microcode Sequencer

5.1 128-Bit Microcode Instruction Format

The accelerator is sequenced by a 64-word $\times$ 128-bit distributed configuration RAM (`config_mem.v`). Each 128-bit instruction fully encodes layer parameters, as detailed in Table 1.

5.2 Multi-Model Workload Mapping

5.2.1 MNIST LeNet-5 (Digit Classifier)

- **Complexity**: 11 uops, 32 KB weights, 1 spatial chunk (32×32 canvas).
- **SRAM Weight Preload**: Prior to execution, 2,048 beats (32 KB in INT8) or 4,096 beats (64 KB in INT16) are streamed from DDR4 into SRAM Rows 0–127 via Read DMA. The sequencer executes with `wt_ddr_phys_addr = 0x0000_0000`, running entirely out of on-chip memory.
- **Performance**: Executes in 1,823 clock cycles (24.31 μs at 75 MHz), achieving $> 41,000$ FPS throughput.

5.2.2 ResNet-50 Full Backbone (1,000 Classes)

- **Complexity**: 57 uops, 23.5 MB (INT8) / 47.1 MB (INT16), 196 spatial chunks (14×14 grid).
- **Execution Flow**: Evaluates all 53 convolutional stages and residual skip connections, concluding with a 4-pass GEMV classification head writing 1,000 logits to DDR.
- **Performance**: 11,414,000 clock cycles (152.19 ms at 75 MHz, 6.6 FPS).

5.2.3 MobileNetV2 (Inverted Residuals)

- **Complexity**: 56 uops, 2.27 MB (INT8) / 4.54 MB (INT16).
- **Depthwise Acceleration**: Utilizes controller Mode 3 (`mode = 2'b11`) to process 3×3 depthwise layers without cross-channel reduction stalls.

5.2.4 EfficientNet-B0 (MBConv + LUT Swish)

- **Complexity**: 53 uops, 3.41 MB (INT8) / 6.82 MB (INT16).
- **Swish & SE Scaling**: Evaluates Swish activations via the on-chip 256-entry lookup table (`act_lut_rom.v`) and executes Squeeze-and-Excitation channel gating via microcode sequencing.

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6 Dedicated DDR Memory Map & Baremetal Stack

6.1 DDR4 Physical Address Allocation

To prevent data corruption during sequential or standalone model evaluation, DDR4 memory is partitioned into dedicated non-overlapping regions:

6.2 Baremetal Control Mailbox Mechanism

A 4-byte mailbox word at `0x21FF_0000` enables dynamic selection of target workloads:

- `0x00`: Execute all 4 models sequentially (comprehensive verification).
- `0x01`: Execute MNIST LeNet-5 only.
- `0x02`: Execute ResNet-50 Full Backbone only.
- `0x03`: Execute MobileNetV2 only.
- `0x04`: Execute EfficientNet-B0 only.

6.3 ARM Cortex-A53 Driver Implementation

The standalone driver (`c_engine_driver.c`) manages hardware interactions:

- **Cache Coherency**: Executes `Xil_DCacheFlushRange` before DMA initiation and `Xil_DCacheInvalidateRange` prior to reading output logits, preventing stale data reads on ARM Cortex-A53.
- **Performance Monitoring**: Reads 64-bit cycle counters from registers `0x48` (`CYCLES_LO`) and `0x4C` (`CYCLES_HI`) to calculate hardware latency and throughput (FPS).
- **Top-K Softmax**: Applies numerically stable Softmax over output logits to extract and display Top-5 predictions on the UART terminal.

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7 Physical Implementation & Timing Closure

7.1 Implementation Configuration

Physical synthesis and implementation were carried out using AMD Vivado 2023.2 targeting the `xczu7ev-ffvc1156-2-e` device on the ZCU106 Evaluation Board.

TABLE 1
128-Bit Microcode Instruction Word Bitfield Allocation

Word	Bit Range	Parameter Description
Word 0 [31:0]	$[1:0]$	Mode (0: Conv2D, 1: Dense GEMV, 2: Pool, 3: Depthwise)
	$[9:2]$	Input Channel Tile Count ($C_{\text{in}}/16$)
	$[17:10]$	Output Channel Tile Count ($C_{\text{out}}/16$)
	$[21:18]$	Kernel Width (K_w)
	$[25:22]$	Kernel Height (K_h)
	$[30:26]$	Requantization Shift Bits ($0 \leq$ shift ≤ 31)
	$[31]$	Padding Type (0: Valid, 1: Same)
Word 1 [63:32]	$[45:32]$	14-Bit SRAM Input Activation Base Row Address
	$[59:46]$	14-Bit SRAM Weight Base Row Address
	$[61:60]$	Activation Type (0: None, 1: ReLU, 2: Leaky ReLU, 3: LUT Swish)
	$[63:62]$	Stride (1, 2, 3)
Word 2 [95:64]	$[77:64]$	14-Bit SRAM Output Activation Base Row Address
	$[91:78]$	14-Bit SRAM Residual Skip Base Row Address
	$[92]$	Residual Addition Enable (`residual_en`)
	$[93]$	Inbound DMA Read Trigger (`dma_rd_en`)
	$[94]$	Outbound DMA Write Trigger (`dma_wr_en`)
	$[95]$	Last Microcode Instruction Flag (`is_last`)
Word 3 [127:96]	$[97:96]$	Pooling Type (0: None, 1: MaxPool, 2: AvgPool)
	$[99:98]$	Pooling Window Size (0: 1x1, 1: 2x2, 2: 3x3)
	$[101:100]$	Pooling Stride
	$[109:102]$	Spatial Chunk Count (1 for MNIST, 196 for ImageNet)
	$[116:110]$	DMA Transfer Burst Length (Beats)

TABLE 2
DDR4 Physical Address Map on AMD ZCU106

Base Address	Size	Target Asset
`0x0000_0000`	512 MB	Baremetal ELF Application
`0x2000_0000`	1 MB	MNIST 32x32 Digit Image (1–2 KB)
`0x2010_0000`	15 MB	ImageNet 224x224 RGB Image (75–150 KB)
`0x2100_0000`	15 MB	Microcode Instructions Buffer
`0x21FF_0000`	4 B	**Model Selection Mailbox**
`0x3000_0000`	16 MB	MNIST LeNet Weights (32–64 KB)
`0x3100_0000`	64 MB	ResNet-50 Weights (23.5–47.1 MB)
`0x3500_0000`	16 MB	MobileNetV2 Weights (2.27–4.54 MB)
`0x3600_0000`	16 MB	EfficientNet-B0 Weights (3.41–6.82 MB)
`0x3800_0000`	1 MB	Output Logits Buffer
`0xA000_0000`	64 KB	AXI-Lite MMIO Registers

7.2 Timing Optimization Strategy

1) **Frequency Scaling for 16-Bit Precision**: In 16-bit mode, the 48-bit accumulator carry chains and wider datapath multiplexers created critical path delays of 14.265 ns. Reducing operating frequency from 75.0 MHz to 70.0 MHz (14.286 ns period) completely eliminated negative slack, achieving positive slack ($WNS = +0.021$ ns) across all 260,579 endpoints.
2) **PPU Pipeline Slicing**: Inserting the pipeline register `data_out_reg` partitioned the critical path from the DSP accumulators into the SRAM write ports into two balanced stages.
3) **LUT-RAM Distributed Sequencer**: Mapping the configuration sequencer into SLICEM distributed RAM eliminated high-fanout routing congestion.

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8 Board-Level Deployment & PowerShell Automation

8.1 Physical Setup

Deploying the system on the physical ZCU106 board involves:

1) Connecting the board via USB-JTAG (`J2`) and USB-UART (`J83`).
2) Configuring DIP switch `SW6` to **JTAG Mode: 0000 (all ON)**.
3) Powering on the board (`SW1`).
4) Opening a serial console (115200 baud, 8-N-1) on the first CP2108 UART interface.

8.2 One-Click PowerShell Commands

Automated deployment scripts in `baremetal/` allow launching evaluations in a single command:

```
# Run all 4 models sequentially
.\baremetal\run_all.ps1 -Precision 8bit -Image dog

# Run MNIST LeNet-5 (Digit 7)
.\baremetal\run_mnist.ps1 -Precision 8bit -Digit 7

# Run ResNet-50 Full Backbone
.\baremetal\run_resnet50.ps1 -Precision 8bit -Image dog

# Run MobileNetV2 (Depthwise Mode = 2'b11)
.\baremetal\run_mobilenetv2.ps1 -Precision 8bit -Image dog

# Run EfficientNet-B0 (MBConv + LUT Swish)
.\baremetal\run_efficientnetb0.ps1 -Precision 8bit -Image
    dog
```

8.3 Sample Execution Output

Listing 1 illustrates the UART console output during sequential multi-model execution.

TABLE 3
Comprehensive Physical Implementation and Resource Utilization Metrics on AMD ZCU106

Resource / Timing Metric	8-Bit Mode (INT8)	16-Bit Mode (INT16)	ZCU106 Capacity	Utilization (%)
Target Clock Frequency (F_{clk})	**75.0 MHz**	**70.0 MHz**	–	–
Clock Period (T_{clk})	13.334 ns	14.286 ns	–	–
Worst Negative Slack (WNS)	**+1.473 ns (MET)**	**+0.021 ns (MET)**	$WNS \geq 0.0$	**100% Clean Closure**
Total Negative Slack (TNS)	**0.000 ns (MET)**	**0.000 ns (MET)**	$TNS = 0.0$	**Zero Violations**
Failing Endpoints	**0 / 221,438**	**0 / 260,579**	0	**Zero Violations**
Worst Hold Slack (WHS)	**+0.013 ns (MET)**	**+0.017 ns (MET)**	$WHS \geq 0.0$	Hold Met
UltraRAM (URAM288) Tiles	**64**	**96**	96	**66.67% / 100.0%**
Block RAM (RAMB36E2) Tiles	**0**	**224**	312	**0.00% / 71.79%**
DSP48E2 Slices	**517**	**517**	1,728	**29.92% / 29.92%**
CLB Look-Up Tables (LUTs)	**74,937**	**131,290**	230,400	**32.53% / 56.98%**
CLB Flip-Flops (Registers)	**56,557**	**88,598**	460,800	**12.27% / 19.23%**
Peak Compute Throughput	**38.4 GOPS**	**35.8 GOPS**	–	$2 \times 256 \times F_{\text{clk}}$

```
=======================================
  AMD ZCU106 UNIFIED C-ENGINE BAREMETAL
  Processing: 256 PEs @ 75.0 MHz (INT8)
  Memory: 64 URAMs (1.0 MB) + 4 GB DDR4
=======================================

[*] Driver initialized at 75.0 MHz.
[*] Target: All 4 Models Sequentially

[1/4] Running MNIST LeNet-5 (10 classes)...
[*] Programming microcode (11 uops)...
[*] Preloading 2048 beats to SRAM...
[+] Weights resident in on-chip SRAM.
[*] Streaming input digit (64 beats)...
[*] Triggering Systolic Compute Engine...
[+] Execution Completed Successfully!
  Cycles:     1823 cyc @ 75.0 MHz
  Latency:    24.31 us (0.024 ms)
  Throughput: 41141.0 FPS
  Top-1: Class #7 (Digit 7) -> 99.87%

[2/4] Running Full ResNet-50 (1000 classes)...
[*] Programming microcode (228 uops)...
[*] Streaming input image (4704 beats)...
[*] Triggering Systolic Compute Engine...
[+] Execution Completed Successfully!
  Cycles:     11414000 cyc @ 75.0 MHz
  Latency:    152.19 ms
  Throughput: 6.6 FPS
  Top-1: Class #207 (golden retriever) 87.4%
  Top-2: Class #153 (Maltese dog)      8.2%
  Top-3: Class #263 (Pembroke corgi)   2.3%
=======================================
```

Listing 1. UART Console Output from AMD ZCU106 Multi-Model Inference.

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9 Conclusion

This paper presented the comprehensive architecture, RTL microarchitecture, DDR tiling methodology, and physical deployment of a unified 256-PE systolic neural accelerator on the AMD Zynq UltraScale+ ZCU106 Evaluation Kit. By integrating dual-precision arithmetic (INT8 @ 75.0 MHz and INT16 @ 70.0 MHz), native depthwise separable convolution (Mode 3), an on-chip activation LUT for Swish/SiLU, and an autonomous 128-bit microcode sequencer, the accelerator achieves complete end-to-end PL hardware execution across MNIST LeNet-5, ResNet-50, MobileNetV2, and EfficientNet-B0 with zero CPU convolution fallback. Guaranteed physical timing closure was demonstrated across all 260,579 endpoints on the ZCU106. The standalone ARM Cortex-A53 baremetal software stack, dedicated non-overlapping DDR4 memory layout, on-chip SRAM weight preloading, and one-click PowerShell execution scripts provide a reliable foundation for edge neural acceleration.

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